

1MT495 Current Issues in Applied International Economics: Foreign Exchange Markets

Martin Casta

Prague University of Economics and Business

November 28, 2025

Outline

- 1 Foreign Exchange Markets
- 2 FX Quotations
- 3 Arbitrage Example
- 4 Exchange Rate Regimes and Policies

Section I: FX Market Structure & Quotation

- Global FX Market Structure & Scale
- FX Trading Instruments and Participants
- Exchange Rate Terminology and Quotation
- Spot FX Arithmetic (Arbitrage Example)

Characteristics: A Deeply Decentralized Market

- The foreign exchange market (FX, Forex) is a segment of the financial market where foreign currencies are traded.
- It is a **highly decentralized, Over-the-Counter (OTC)** market.
- **Near-24-hour operation:** Trading occurs almost 24 hours a day, 5 days a week (no trading during weekends).

Forex session		Local time	Singapore Time (SST)	Coordinated Universal Time (UTC)	Eastern Standard Time (EST)
Sydney	Open	08:00	08:00	21:00	17:00
	Close	16:00	14:00	06:00	01:00
Tokyo	Open	09:00	08:00	00:00	19:00
	Close	18:00	17:00	09:00	04:00
London	Open	08:00	16:00	08:00	03:00
	Close	16:00	00:00	16:00	11:00
New York	Open	08:00	21:00	13:00	08:00
	Close	17:00	06:00	22:00	17:00

Source: BIS

- **High Liquidity & Efficiency:** A highly liquid market, very close to a **perfectly competitive market** due to minimal barriers to entry and vast participant size.

Volume: The BIS Triennial Survey

- Measuring trading volume is difficult due to the OTC and decentralized nature of the market.
- The **Bank for International Settlements (BIS) Triennial Survey** is the industry standard for volume measurement.
- **Average Daily Turnover (April 2022): \$7.5 Trillion USD** (Update from 2020's \$6.6T).
- **Transaction Breakdown (2022):**
 - Spot Transactions: ~ 28% (T+2 settlement).
 - FX Swaps: ~ 51% (Largest share, used for hedging and funding).
 - Forwards, Options, and other: ~ 21%.
- **For more information see:**
 -
 - BIS article: *The foreign exchange market* (2023)
 - *Lehman Brothers Foreign Exchange Training Manual*

BIS: Most Traded Currency Pairs

(%)	April 2001	April 2004	April 2007	April 2010	April 2013	April 2016	April 2019
USD/EUR	30	28	27	28	24	23	24
USD/GBP	11	14	12	9	9	9	10
USD/JPY	20	17	13	14	18	18	13
USD/CHF	5	4	5	4	3	4	4
USD/CAD	4	4	4	5	4	4	4
USD/AUD	4	5	6	6	7	5	5
USD/oth. .	17	18	19	11	10	12	35
EUR/JPY	3	3	2	3	3	2	2
EUR/GBP	2	2	2	3	2	2	2
EUR/CHF	1	1	2	2	1	1	1
EUR/oth. .	2	2	4	3	3	3	3

Key Takeaway: The **USD** is involved in nearly 90% of all transactions, affirming its status as the world's dominant vehicle currency.

BIS: Daily FX Turnover

Bil.\$	1998	2001	2004	2007	2010	2013	2016	2019
Great Britain	685	542	835	1 483	1 854	2 726	2 406	3 576
USA	383	273	499	745	904	1 263	1 272	1 370
Japan	146	153	207	250	312	374	399	376
Singapore	145	104	134	242	266	383	517	640
Hong Kong	80	68	106	181	238	275	437	632
Poland	3	5	7	9	8	8	9	9
Czech republic	5	2	2	5	5	5	4	7
Hungary	1	1	3	7	4	4	3	4
Slovakia	0	1	2	3	0	1	2	1

Source: BIS

Key Participants and Roles

- **Dealers / Market Makers (MMs):** Financial institutions (mostly banks) who buy/sell FX on their own account and risk. They quote two-way prices (Bid and Ask).
- **Brokers:** Intermediaries (increasingly electronic via platforms like EBS and Reuters) that connect buyers and sellers, earning a fee.
- **Speculators & Arbitrageurs:** Individuals or funds seeking profit from expected price movements (speculators) or price discrepancies (arbitrageurs).
- **Central Banks:** Act as dealers (for official reserve management) or intervene for policy reasons.
- **Corporations and Individuals:** Transacting for international trade, investment, or tourism.

Core FX Trading Motivations

- **Arbitrage:** Earning profit from simultaneous price discrepancies (e.g., cross-rate differences).
 - **Characteristics:** Risk-free; enforces equilibrium conditions like the **Covered Interest Parity (CIP)**.
- **Speculation:** Earning profit from an expected favorable future price change.
 - **Characteristics:** High risk; driven by expectations related to the **Uncovered Interest Parity (UIP)** hypothesis.
- **Hedging:** Taking an offsetting position to reduce exposure to exchange rate risk (e.g., by using forwards or options).
- **Transaction Demand:** Exchanging currencies to facilitate international trade or investment.

Execution Techniques (Instruments)

- **Spot:** An agreement to exchange currency now, with physical settlement occurring on the **T+2 business day** (T+1 for some USD/CAD).
- **Forward:** An agreement to exchange a specific amount of currency at a predetermined exchange rate on a specified future date.
- **Future:** A standardized forward contract traded on an exchange.
- **Option:** Gives the buyer the right (but not the obligation) to buy or sell currency at a set rate (strike price) on or before a future date.
- **FX Swap:** A simultaneous purchase and sale of the same currency amount with two different value dates (e.g., Spot and Forward).
Crucial for funding and hedging.

Basic Terminology

- **Exchange Rate (E):** Price of one currency (Base) in units of another (Counter).
- **Appreciation (Strengthening):** increase of value E due to market forces (Floating regimes).
- **Depreciation (Weakening):** decrease of value E due to market forces (Floating regimes).
- **Revaluation:** An **official, discrete increase** of the central parity by the government/CB (Fixed regimes only).
- **Devaluation:** An **official, discrete decrease** of the central parity by the government/CB (Fixed regimes only).

Key Distinction: Market forces drive appreciation/depreciation. Policy decisions drive revaluation/devaluation.

Base and Counter Currency Conventions

- **Base Currency (Unit Currency):** The currency whose value is fixed at one unit (denominator, left).
- **Counter Currency (Quote Currency):** The currency whose amount varies (numerator, right).

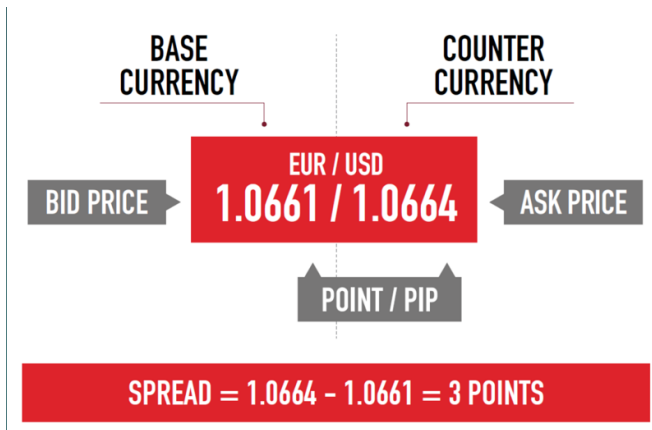
Standard Market Conventions (Base Currency):

- 1 Generally: **EUR, GBP, AUD, NZD, USD, CAD, CHF, JPY** (in that order of precedence, if two are involved).
- 2 Exceptionally: For most currency pairs involving a local currency (CZK, TRY, etc.), the **USD** or **EUR** is the standard Base Currency (e.g., USD/CZK or EUR/CZK).

Example: EUR/USD = 1.0800

Here, **EUR** is the Base currency. 1 EUR costs 1.0800 USD.

Quotation



Appreciation vs. Depreciation: The Importance of Quotation

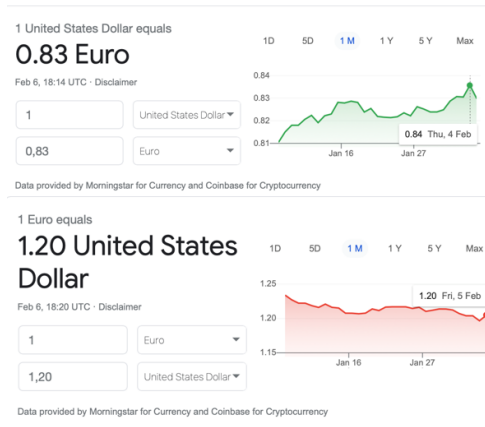
- Whether a numerical increase in the exchange rate means appreciation or depreciation depends entirely on which currency is the Base currency.

Case Study: USD/CZK rises from 22.00 to 25.00

- Numerical value **rises**.
- It now costs **more CZK** (25.00) to buy 1 USD.
- **Conclusion:** The **CZK has depreciated** (weakened) against the USD. The **USD has appreciated** (strengthened) against the CZK.

General Rule: A numerical increase in $E_{\text{Base/Counter}}$ means the **Base** currency has **appreciated** against the **Counter** currency.

Depreciation and Appreciation



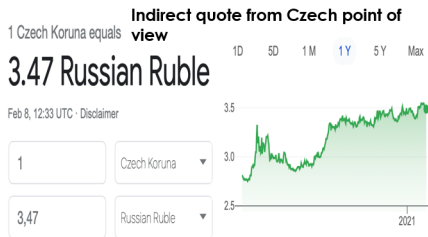
On the top we have a USD/EUR, we can see the appreciation of dollar in relation to euro (by the beginning of the period, one dollar could get you 0.81 eurocents). But when the exchange rate is recorded as EUR/USD, it decreases while its appreciation of dollar and depreciation of euro

Direct vs Indirect Quotation

Direct Quote (Domestic Perspective): $\frac{\text{Domestic Currency}}{1 \text{ Foreign Currency}}$
(e.g., CZK 25.00 per 1 USD)

Indirect Quote (Domestic Perspective): $\frac{1 \text{ Domestic Currency}}{\text{Foreign Currency}}$
(e.g., USD 0.04 per 1 CZK)

Direct and indirect quotation



Bid-Ask Spread and Profit

- **Bid:** The rate at which the dealer is willing to ****Buy**** the Base currency (lower rate).
- **Ask (Offer):** The rate at which the dealer is willing to ****Sell**** the Base currency (higher rate).
- The difference (**Ask - Bid**) is the **Bid-Ask Spread**, representing the dealer's profit margin.
- **Pip (Point in Percentage):** The smallest unit of movement (e.g., 0.0001 for most pairs).

$$\text{Spread (\%)} = \frac{\text{Ask} - \text{Bid}}{\text{Ask}} \times 100$$

Factors Affecting Spread: Volume, Volatility, Currency Liquidity/Credibility.

Quotation

Currency	Country	Quantity	We buy ?	We sell ?	Mid
Favourite currencies					
 GBP - British pound	Great Britain	1	28,825	30,304	29,564
 EUR - Euro	EU	1	25,253	26,548	25,900
 USD - United States dollar	USA	1	21,091	22,173	21,632
Other currencies					
 AUD - Australian dollar	Australia	1	16,036	16,858	16,447
 BGN - Bulgarian lev	Bulgaria	1	12,912	13,574	13,243
 CAD - Canadian dollar	Canada	1	16,440	17,283	16,862
 HRK - Croatian kuna	Croatia	1	3,335	3,506	3,421
 DKK - Danish krone	Denmark	1	3,396	3,570	3,483

The quotations such as in the table (i.e., the quotations consisting of complete entry of the exchange rate) are called outright.

Bilateral Arbitrage: Exploiting Dealer Spreads

Goal: Start with \$1,085,500 USD and find a risk-free profit by trading EUR/USD between two dealers.

The Setup (EUR is Base):

- **Dealer A (Ask):** 1.0853
- **Dealer B (Bid):** 1.0855

Arbitrage Condition: Dealer A's Ask (selling EUR) is lower than Dealer B's Bid (buying EUR).

$$1.0853 < 1.0855$$

Risk-Free Profit:

$$\text{\$1,085,720.00} - \text{\$1,085,500.00} = \text{\$220.00}$$

The Cycle (Start \$1,085,500 USD):

- 1 Buy EUR from Dealer A (Use Ask 1.0853):

$$\frac{\text{\$1,085,500}}{1.0853/\text{EUR}} = 1,000,184.28 \text{ EUR}$$

- 2 Sell EUR to Dealer B (Use Bid 1.0855):

$$1,000,184.28 \text{ EUR} \times \$1.0855/\text{EUR} = \text{\$1,085,720.00 US}$$

Multilateral (Triangular) Arbitrage

- Involves three currencies and three exchange rates.
- Arbitrage exists if the **cross rate** implied by two pairs differs from the direct quote of the third pair.
- Profit is generated by cycling funds through the three pairs and ending up with more of the starting currency.

Two possible cycles:

- ① USD $\Rightarrow$ EUR $\Rightarrow$ CZK $\Rightarrow$ USD
- ② USD $\Rightarrow$ CZK $\Rightarrow$ EUR $\Rightarrow$ USD

Example Cycle: Start with USD, buy EUR, use EUR to buy CZK, use CZK to buy back USD.

Arbitrage Exists If: $E_{\text{USD}/\text{EUR}} \times E_{\text{EUR}/\text{CZK}} \neq E_{\text{USD}/\text{CZK}}$

Triangular Arbitrage Example: The Market Setup

Goal: Start with \$1,000,000 USD and identify a misalignment in the USD-JPY-CAD triangle. (Assume mid-rates for simplicity).

Table: Market Quotes for Three Currency Pairs

Dealer / Pair	Quote	Base Currency
Dealer 1: $E_{\text{CAD/USD}}$	0.7300	USD
Dealer 2: $E_{\text{JPY/USD}}$	148.00	USD
Dealer 3: $E_{\text{CAD/JPY}}$	0.004900	JPY

The Key Question: Does the direct market rate ($E_{\text{CAD/JPY}}$) equal the rate implied by the two USD crosses ($E_{\text{CAD/USD}}$ and $E_{\text{JPY/USD}}$)?

Step 1: Calculate the Implied No-Arbitrage Cross Rate

The **No-Arbitrage Cross Rate** for CAD/JPY is derived from the two USD quotes.

- **Formula:** The implied rate must be consistent:

$$E_{\text{CAD/JPY}}^{\text{Implied}} = \frac{E_{\text{CAD/USD}}}{E_{\text{JPY/USD}}}$$

- **Calculation:**

$$E_{\text{CAD/JPY}}^{\text{Implied}} = \frac{0.7300 \text{ CAD/USD}}{148.00 \text{ JPY/USD}} \approx \mathbf{0.004932}$$

Discrepancy Check:

- Market Quote ($E_{\text{CAD/JPY}}^{\text{Market}}$): 0.004900
- Implied Rate ($E_{\text{CAD/JPY}}^{\text{Implied}}$): 0.004932

Conclusion: Since $0.004900 < 0.004932$, the market is willing to sell JPY for CAD too cheaply. The profitable cycle must involve **selling the relatively cheap JPY** and **buying the relatively expensive JPY** via the cross rate.

Step 2: Execute the Profitable Cycle (\$1,000,000 USD)

The cycle must be: **USD $\Rightarrow$ CAD $\Rightarrow$ JPY $\Rightarrow$ USD.**

① **Convert USD $\rightarrow$ CAD (Dealer 1):**

$$1,000,000 \text{ USD} \times 0.7300 \text{ CAD/USD} = \mathbf{730,000 \text{ CAD}}$$

② **Convert CAD $\rightarrow$ JPY (Dealer 3):** We need

$$E_{\text{JPY/CAD}} = 1/E_{\text{CAD/JPY}} = 1/0.004900 \approx 204.0816$$

$$730,000 \text{ CAD} \times 204.0816 \text{ JPY/CAD} = \mathbf{148,979,591.8 \text{ JPY}}$$

③ **Convert JPY $\rightarrow$ USD (Dealer 2):** We need

$$E_{\text{USD/JPY}} = 1/E_{\text{JPY/USD}} = 1/148.00 \approx 0.0067567$$

$$148,979,591.8 \text{ JPY} \times 0.0067567 \text{ USD/JPY} = \mathbf{\$1,006,618.86 \text{ USD}}$$

Step 3: Calculate the Arbitrage Profit

- **Starting Capital:** \$1,000,000.00 USD
- **Ending Capital:** \$1,006,618.86 USD

Risk-Free Profit Calculation:

$$\text{Profit} = \text{Ending Capital} - \text{Starting Capital}$$

$$\text{Profit} = \$1,006,618.86 - \$1,000,000.00 = \mathbf{\$6,618.86}$$

Economic Implication: This profit opportunity is immediately seized by High-Frequency Trading (HFT) algorithms, ensuring the triangular parity holds in deep, liquid markets like FX.

Recap: Key Takeaways (Section I)

- FX markets are a **global OTC** network, dominated by **FX Swaps** in daily turnover.
- **Exchange Rates** are quotes (Base/Counter) that determine appreciation/depreciation.
- **Appreciation/Depreciation** is market-driven;
Revaluation/Devaluation is policy-driven.
- Trading motivations are **Transaction, Hedging, Speculation**, and **Arbitrage**.
- **Arbitrage** is risk-free and enforces market equilibrium (e.g., CIP).

Exchange Rate Policy & Convertibility

- Exchange Rate Policy involves two main components:
 - ① **Choice of Exchange Rate Regime** (e.g., fixed vs. floating).
 - ② **Exchange Rate Management** (e.g., policy interventions).
- **Convertibility:** The ability of residents and non-residents to freely exchange currency (a prerequisite for participation in deep FX markets).
- **Capital Controls:** Limitations on convertibility that may cause dual exchange rates or centralized FX trading (e.g., mandatory sale of export revenues).

De Jure Regimes: IMF Classification Spectrum

The IMF classifies regimes on a spectrum of flexibility:

- ① **Hard Pegs:** The most rigid commitment. Monetary policy is entirely sacrificed.
 - Dollarization (e.g., Panama)
 - Currency Board (e.g., Hong Kong, Bulgaria)
- ② **Soft Pegs:** Pegged within a band, or adjustable/crawling. Requires frequent FX intervention.
- ③ **Floating:** Determined by supply and demand.
 - Managed Floating (interventions allowed, e.g., Czech Republic, Korea)
 - Free Floating (no or minimal interventions, e.g., US, UK, Japan)

Exchange Rate Regimes

Flexible regime:

- ER determined by supply and demand.

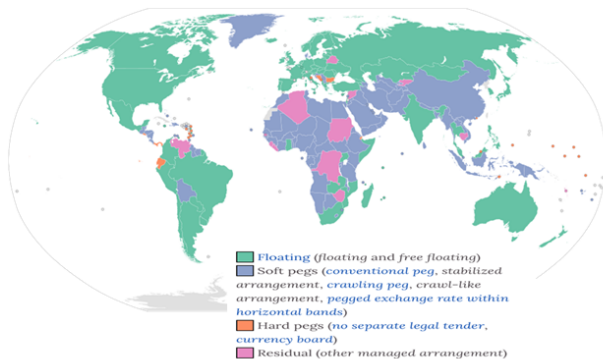
Fixed regime:

- ER stabilized/pegged to a currency or band.
- Requires FX interventions.

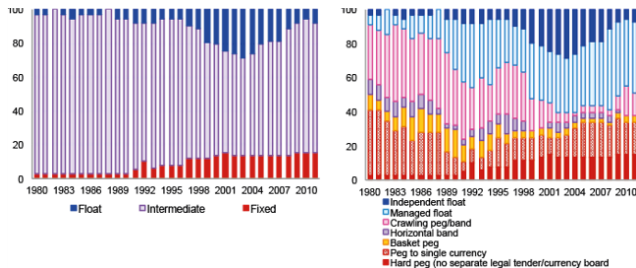
Regime	Description
Dollarization, euroization	Lacks a separate legal institution
Currency board arrangement	The money base is fully backed by an equal amount of foreign exchange reserves
Conventional fixed peg	Peg to some foreign currency or currency basket with a maximum deviation (bands) of $\pm 1\%$
Horizontal band	Peg with bands larger than $\pm 1\%$
Crawling peg	Fixed rate which is revised, if necessary, according to changes in selected indicators
Crawling band	Crawling peg combined with bands larger than $\pm 1\%$
Managed float	Floating exchange rate with active intervention
Independent float	A rate determined by market forces; monetary policy is independent of exchange rate policy

Source: Exchange Rate Arrangements (1999).

De Jure Exchange Rate Regimes (2019, de jure IMF classifications)



De Jure Exchange Rate Regimes



De Jure vs De Facto

Problem: official regime often differs from actual behaviour.

This creates **exchange rate regime gaps**.

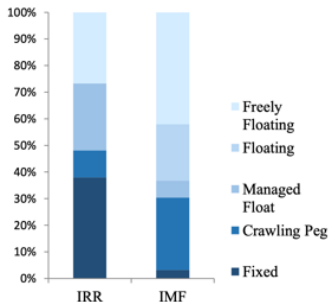
Two main approaches for De Facto Classifications:

- **Ilzetzki, Reinhart & Rogoff**: policy actions, black-market ERs.
- **Levy-Yeyati & Sturzenegger**: volatility of ER and reserves.

De Facto Classifications

Ilzetzki, Reinhart & Rogoff (2017)

Figure 8: Share of World GDP by Exchange Arrangement: Ilzetzki, Reinhart and Rogoff (2016) and IMF (2014)



De Jure vs. De Facto Regimes (The Gap)

Problem: Official (De Jure) regime often differs significantly from actual policy behavior (De Facto). This leads to **Exchange Rate Regime Gaps** (Calvo & Reinhart, 2002).

- **Fear of Floating (The most common gap):** A country declares a flexible (floating) rate, but the central bank frequently intervenes to dampen exchange rate volatility, behaving like a soft peg.
- **Fear of Pegging (Rare):** A country declares a fixed peg but allows the exchange rate to fluctuate freely.
- **Fear of Free Falling:** The extreme case of defending an unsustainable peg, leading to reserve depletion (e.g., during a balance of payments crisis).

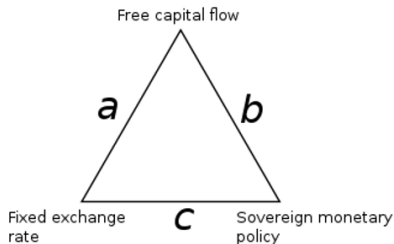
Exchange Rate Regime Choice

- **Fixed:** good for inflation reduction, but risky in crises.
- **Floating:** flexible and adaptive.
- **Managed:** hybrid with discretionary interventions.

The Impossible Trinity (Trilemma, Mundell-Fleming)

A country cannot simultaneously maintain all three of the following policy goals. It must choose two and sacrifice the third:

- ① **Fixed Exchange Rate**
(Credibility/Stability)
- ② **Free Capital Flows**
(Efficiency/Growth)
- ③ **Independent Monetary Policy**
(Autonomy/Stability)



Most developed nations choose (2) and (3), resulting in floating rates.